



**PROJECT SYNOPSIS**

**ON**

**BLOOD BANK & EMERGENCY DONOR CONTACT**

Submitted for

Database Management Systems

Submitted By

Dhruv Mittal (1024030364)

Vishaw Bansal (1024030356)

Sahil Mangla (1024030359)

**Submitted to**

**Dr. Amrita Dahiya**

## ABSTRACT

In an emergency, time is everything. Imagine a situation where a patient urgently needs blood, but finding a matching donor becomes a race against time. The Blood Bank & Emergency Donor Connect system is designed to solve this critical problem by providing a centralized and efficient platform to connect donors with patients instantly.

This database-driven application maintains structured and real-time information about blood donors, including their availability, blood group, and location. By leveraging DBMS concepts such as entity relationships, normalization, and optimized querying, the system ensures fast, reliable, and accurate access to donor data when it matters the most.

## 1. Introduction

Every year, thousands of lives are at risk not because treatment is unavailable, but because **blood is not found on time**. In many cases, families struggle to locate donors, relying on personal contacts, social media, or random calls—all of which are slow and unreliable.

This project aims to change that. Instead of chaos, it introduces a **structured, database-driven system** where donor information is organized, searchable, and instantly accessible.

The goal is simple: *Reduce the time between need and help.*

## 2. Problem Statement

Finding blood during emergencies is often unorganized and stressful. Existing methods:

- Depend heavily on manual efforts
- Lack real-time donor availability
- Contain inconsistent or duplicate data
- Fail to provide quick and reliable search results

As a result, critical time is lost — and in emergencies, that delay can cost lives.

### 3. Objectives of the Project

The primary objective of this project is to build a system that reduces the time and effort required to find blood donors during emergencies. Instead of relying on manual searches and uncertain sources, the system aims to provide **quick, reliable, and structured access to donor information**.

Key objectives include:

- To create a centralized and well-structured database of blood donors
- To enable fast and accurate search based on blood group and location
- To maintain real-time donor availability for better reliability
- To ensure data integrity and minimize redundancy using normalization
- To provide an efficient connection between donors and recipients in emergency situations
- To optimize data retrieval using efficient queries and indexing

### 4. Motivation Behind the Project:

This project is not just about databases — it is about impact.

Behind every blood request is a real person: a patient in surgery, a victim of an accident, or someone fighting for survival.

The motivation is to use technology to ensure that **no one suffers due to lack of timely information**. Even a few minutes saved in finding a donor can make a life-saving difference.

### 5. Scope of the Project

The scope of this project focuses on building a practical and efficient system that can manage blood donor information and assist users during emergency situations.

The system includes the following functionalities:

- Registration and management of donor profiles
- Storage of essential details such as blood group, location, and availability
- Quick search functionality to find suitable donors based on urgency
- Ability for users to raise blood requests during emergencies
- Real-time updating of donor availability status

The project is designed primarily for **academic demonstration of DBMS concepts**, including data modeling, normalization, and efficient querying.

## 6. System Description

The system works like a bridge between **those who need blood and those willing to donate**.

- A user logs in and raises a request
- The system instantly searches the database
- Matching donors are identified based on blood group and location
- Relevant results are displayed within seconds

Behind this simple experience lies a well-structured backend powered by efficient database operations and optimized queries.

## 7. Module Description

### 1. User Authentication Module

Ensures that only verified users can access the system, maintaining security and data integrity.

### 2. Donor Module

Allows individuals to register as donors and update their availability, making the system dynamic and reliable.

### 3. Request Module

Enables users to quickly raise emergency blood requests with necessary details.

### 4. Search Module

Acts as the core of the system — instantly filtering and displaying the most relevant donors based on urgency.

## 8. Database Design

The system is built on a structured database that efficiently stores and manages donor and request information. The design focuses on **clarity, scalability, and fast data retrieval**.

### Main Entities:

#### Donor

- DonorID (Primary Key)
- Name
- BloodGroup
- Location
- ContactDetails
- AvailabilityStatus

## User

- UserID (Primary Key)
- Name
- Email
- Password

## BloodRequest

- RequestID (Primary Key)
- UserID (Foreign Key)
- BloodGroup
- Location
- Status (Pending/Completed)

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## Relationships:

- A **User** can create multiple **Blood Requests** (One-to-Many)
- A **Donor** can fulfill multiple requests over time (Many-to-Many conceptually)

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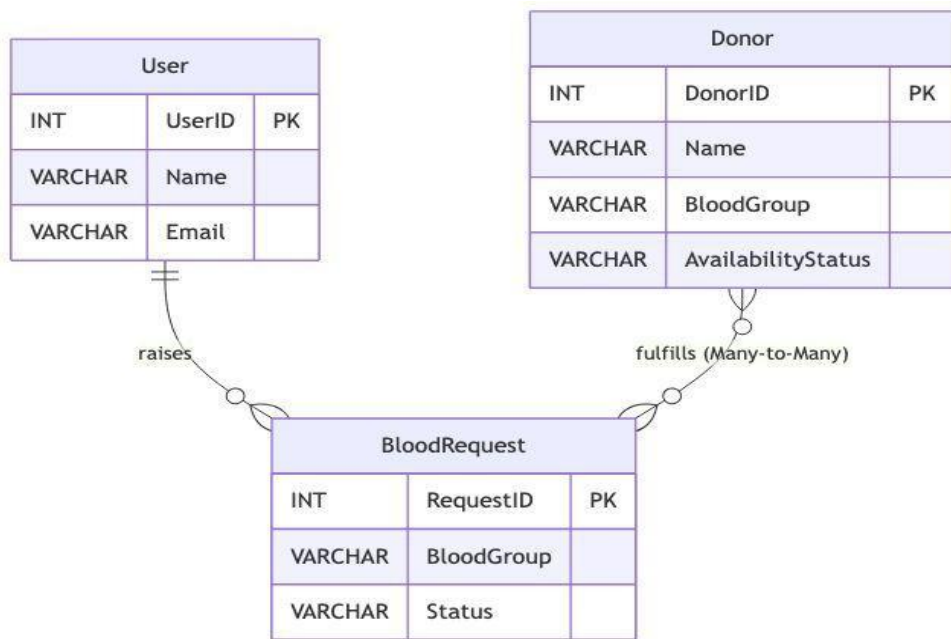
## Design Considerations:

- Indexing on **BloodGroup** and **Location** for faster search
- Use of unique identifiers (IDs) to maintain entity distinction
- Proper referencing using foreign keys to ensure relational integrity

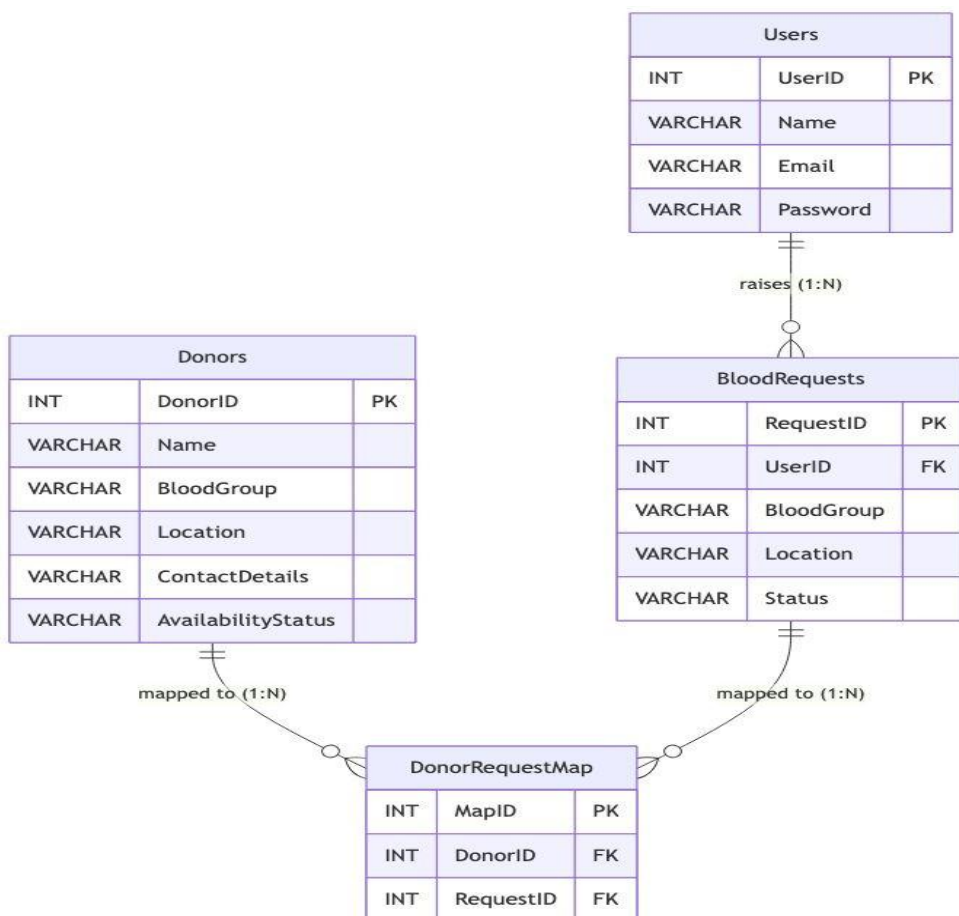
This design ensures that the system remains **efficient, scalable, and easy to maintain**.

## 9. ER Diagram Description

The Entity–Relationship (ER) diagram represents the core entities of the system, namely **User**, **Donor**, and **BloodRequest**, along with their relationships. A one-to-many relationship exists between users and blood requests, as a single user can raise multiple requests. The system also logically connects donors with requests based on blood group and location. The ER model provides a clear and structured representation of the database, ensuring efficient organization, retrieval, and management of data.



(ER Model)



(Relational Model)

## 10. Normalisation

To ensure efficient data management, the database is normalized up to **Third Normal Form (3NF)**.

### Normalization Process:

- **First Normal Form (1NF):**

All attributes contain atomic values with no repeating groups

- **Second Normal Form (2NF):**

All non-key attributes are fully dependent on the primary key

- **Third Normal Form (3NF):**

No transitive dependencies exist; attributes depend only on the primary key

## 11. Technologies Used

The system is developed using modern technologies that ensure efficiency and scalability:

### Database & Backend Logic: Oracle SQL & PL/SQL:

A relational database management system (RDBMS) used for structured data storage and ensuring data integrity. Server-side logic, data manipulation, and automation are handled entirely inside the database using PL/SQL stored procedures, functions, and triggers.

### Tools:

- **Oracle Live SQL / SQL Developer** – Database environment for executing scripts
- **VS Code** – Development environment
- **Git** – Version control and collaboration

## 12. Conclusion

The Blood Bank & Emergency Donor Connect system is more than just a DBMS project — it is a step toward solving a real-world problem using robust database technology.

By combining structured relational data, efficient SQL querying, and automated PL/SQL processing, the system reduces delays and improves response time during emergencies. With further development, this backend architecture has the potential to evolve into a scalable platform that can genuinely save lives.